Key terms used throughout this chapter

data packet – a small part of a message/data that is transmitted over a network; after transmission all the data packets are reassembled to form the original message/data

packet header – the part of the data packet that contains the IP addresses of the sender and receiver, and includes the packet number which allows reassembly of the data packets

packet trailer – the part of a data packet that indicates the end of the data packet and cyclic redundancy check error check

cyclic redundancy check (CRC) – an error checking method in which all the 1-bits in the data packet payload are added and the total is stored in the packet trailer; the same calculation is repeated at the receiving station

payload – the actual data being carried in a data packet

node – stages in a network that can receive and transmit data packets; routers are nodes in communication networks

packet switching – a method of transmission in which a message is broken into many data packets which can then be sent along pathways independently of each other

router – a device that enables data packets to be moved between different networks, for example to join a LAN to a WAN

real time streaming – the transmission of data over a network for live events where the data is sent as soon as it is received or generated

hopping/hop number – a number in a data packet header used to stop data packets that never reach their destination from 'clogging up' the data paths/routes

simplex – data that can be sent on one direction only

half-duplex – data that can be sent in both directions but not at the same time

full-duplex – data that can be sent in both directions at the same time (simultaneously)

serial data transmission – sending data down one channel/wire one bit at a time

parallel data transmission – sending data down several channels/wires several bits at a time (usually 1 byte)

skewed (data) – data that arrives at the destination with the bits no longer synchronised

universal serial bus (USB) – a type of serial data transmission which has become the industry standard for connecting computers to devices via a USB port

parity check – a method used to check if data has been transferred correctly; it makes use of even parity (an even number of 1-bits) or odd parity (an odd number of 1-bits)

parity bit – a bit (either 0 or 1) added to a byte of data in the most significant bit position; this ensures that the byte follows the correct even parity or odd parity protocol

parity block – a horizontal and vertical parity check on a block of data being transmitted

parity byte – an extra byte of data sent at the end of a parity block; it is composed of the parity bits generated from a vertical parity check of the data block

checksum – a verification method used to check if data transferred has been altered or corrupted; calculated from the block of data of data being sent; the checksum value is sent after each data block

automatic repeat request (ARQ) – a method of checking transmitted data for errors; it makes use of acknowledgement and timeout to automatically request re-sending of data if the time interval before positive acknowledgement is too long

acknowledgement – a message sent to the receiver indicating that data has been received correctly (used in the ARQ error detection method)

timeout – the time interval allowed to elapse before an acknowledgement is received (in the ARQ error detection method)

echo check – a method used to check if data has been transferred correctly; data is sent to a receiver and then immediately sent back to the sender; the sender then checks if the received data matches the sent data

check digit – an additional digit appended to a number to check if the entered number is error-free; check digit is a data entry check and not a data transmission check

eavesdropper – another name for a hacker who intercepts data being transmitted on a wired or wireless network

encryption – the process of making data meaningless using encryption keys; without the correct decryption key the data cannot be decoded (unscrambled)

plaintext – the original text/message before it is put through an encryption algorithm

ciphertext – encrypted data that is the result of putting a plaintext message through an encryption algorithm

encryption algorithm – a complex piece of software that takes plaintext and generates an encrypted string known as ciphertext

symmetric encryption – a type of encryption in which the same encryption key is used both to encrypt and decrypt a message

asymmetric encryption – a type of encryption that uses public keys and private keys to ensure data is secure

public key – a type of encryption key that is known to all users

private key – a type of encryption key which is known only to the single computer/user

quantum computer – a computer that can perform very fast calculations; it can perform calculations that are based on probability rather than simple 0 or 1 values; this gives a quantum computer the potential to process considerably more data than existing computers